



Graph Autoencoders for Credit-Card Fraud Detection

Unsupervised learning on graphs and heterogeneous
graph auto-encoders — a critical review

Group 6



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1 Introduction

Why credit-card fraud detection is hard

2 Background

Graph-learning foundations for the model

3 Methodology

How the HGAE works, step by step

4 Experiments

Dataset, protocol, and metrics

5 Results

What the numbers reveal — read critically

6 Conclusion

Takeaways and future directions

SECTION 1

Introduction

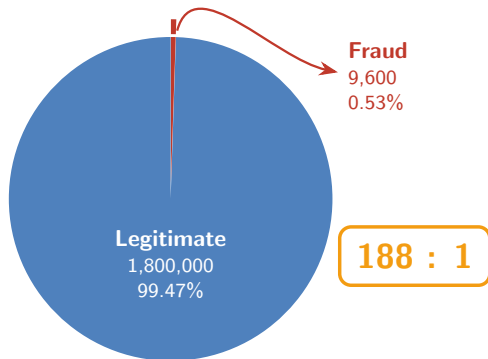
Why detecting credit-card fraud is hard — and why a label-free approach is needed.

Fraud is rare — and that is exactly the problem

- Digital payments exploded → sophisticated fraud
- Supervised ML is the default defence. . .
- . . . but fraud labels are **scarce, delayed, costly**
- Adversaries adapt → yesterday's labels age fast

Key takeaway

Learn what **normal** looks like instead of chasing known fraud.



Problem: extreme imbalance meets relational data

- **188 : 1** imbalance breaks supervised training
- Resampling (SMOTE) → **overfitting** / lost structure
- Transactions are **relational**, not i.i.d. rows

Research question

Can we detect fraud *without fraud labels* by learning the **topology** and **attributes** of legitimate behaviour?

Why existing methods fall short

Supervised / tabular

- Need many labels → infeasible
- Treat each transaction as **isolated**
- Ignore shared cards / merchants / bursts

What we actually need

- Works with **few / no labels**
- Captures **relational** structure
- Stays **scalable** & **interpretable**

Key takeaway

Topology is precisely the signal row-wise classifiers discard.

Objectives & contributions

1. **Survey**: from message-passing GNNs to self-supervised GAEs
2. **Dissect**: the Heterogeneous Graph Auto-Encoder (HGAE)
3. **Evaluate**: architecture, training, reported results — critically
4. **Expose**: formulation & reproducibility gaps (*our contribution*)
5. **Propose**: research agenda for robust detection (*our contribution*)

SECTION 2

Background

The graph-learning foundations needed to understand the model.

HGAE stands on two intellectual lines



Heterogeneous attention

HGT, GEM — type/relation-aware

Reconstruction self-supervision

GAE, VGAE — no labels

Background: message passing, attention, and its limits

- **Message passing**: a node refines itself by aggregating neighbours
- **GAT**: learns *how much* each neighbour matters
- **Permutation equivariance**: the right inductive bias for networks

Caveat

Power bounded by the **1-WL** test; deep stacks cause **oversmoothing**. (*Both return in our critique.*)

The core idea: reconstruction error as an anomaly score

- Encoder compresses graph $\rightarrow$ latent Z
- Decoder reconstructs the input
- Train on **normal** data only
- Anomalies $\rightarrow$ **high reconstruction error**

One-class framing

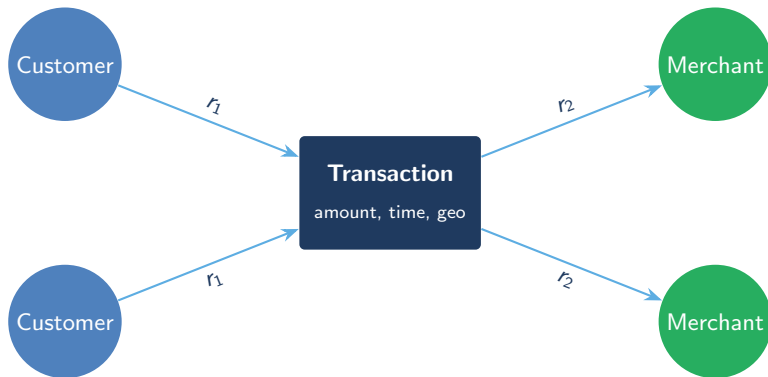
Never having modelled fraud, the decoder rebuilds it *poorly* — the error betrays it.

SECTION 3

Methodology

How the Heterogeneous Graph Auto-Encoder turns transactions into anomaly scores.

Step 1 — Lift transactions into a heterogeneous graph



Three node types, typed edges r_1, r_2

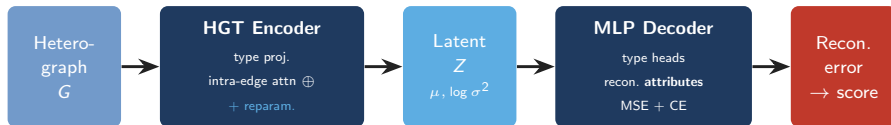
IDs → connectivity only (not features)
is_fraud excluded from training

Step 2 — Encode attributes into a shared space

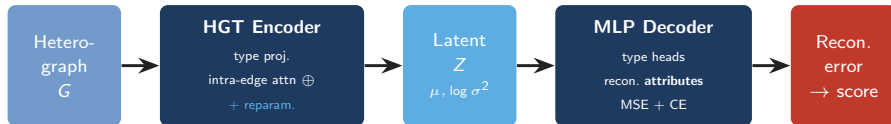
- Continuous (amount, time, geo) → **standardised**
- Categorical (gender, category) → **one-hot**
- Missing → per-type **mean / mode**
- Concatenate ⇒ layer-0 input h^0

Three types ⇒ different dimensions ⇒ reconciled by **type-specific linear projections**.

Step 3 — The HGAE pipeline at a glance

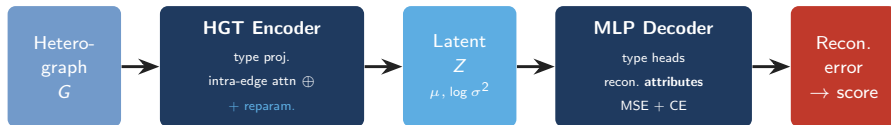


Step 3 — The HGAE pipeline at a glance



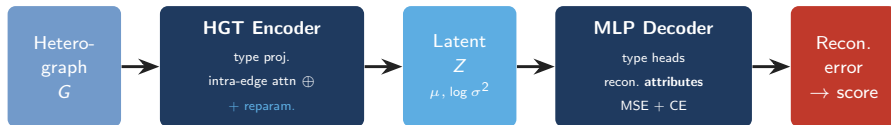
- **Encoder (HGT):** type projections + intra-edge attention + reparameterization

Step 3 — The HGAE pipeline at a glance



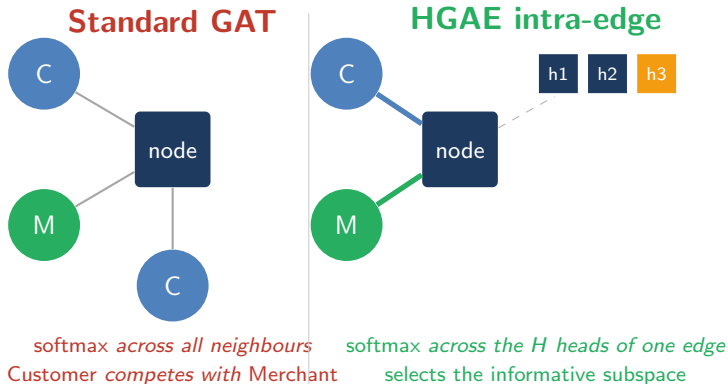
- **Encoder** (HGT): type projections + intra-edge attention + reparameterization
- **Latent Z** : a smooth, probabilistic embedding space

Step 3 — The HGAE pipeline at a glance



- **Encoder** (HGT): type projections + intra-edge attention + reparameterization
- **Latent Z** : a smooth, probabilistic embedding space
- **Decoder** reconstructs **attributes**, *not* adjacency $\rightarrow$ error = score

Key novelty — intra-edge attention respects entity types



$$f_{\text{Attent}} = \text{Softmax}(\|_{k \in [1, H]} \text{Att}^k(v_t, e_r, d_t)) \text{ — normalised over heads, not neighbours}$$

Step 4 — A probabilistic latent space

- Reparameterization at the final encoder layer
- Each embedding = a **Gaussian** (mean, log-var)
- Noise $\rightarrow$ **smooth** space; sampling stays **differentiable**

$$h = \text{mean}(h) + \epsilon \cdot \exp\left(\frac{1}{2} \log(h)\right), \quad \epsilon \sim \mathcal{N}(0, 1)$$

Caveat

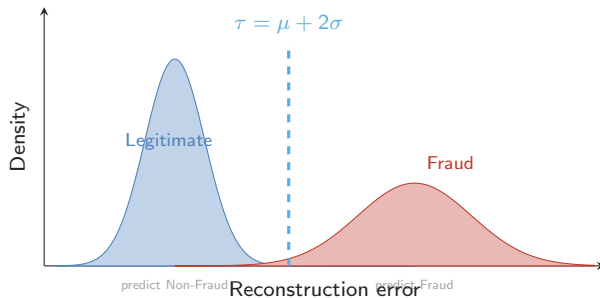
Loss has **no KL term** $\Rightarrow$ strictly a noise-injected AE, not a true VAE.

Step 5 — Decode, then threshold the error

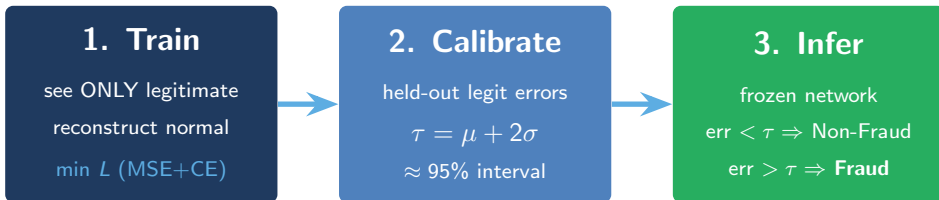
- Type-specific MLP decoder
- err below $\tau \Rightarrow$ **Non-Fraud**
- err above $\tau \Rightarrow$ **Fraud**

$$\tau = \mu + 2\sigma$$

$\approx$ 95% interval of normal error



Step 6 — Three-phase training, no fraud labels in the loss



$$L = \sum_{\forall N} \sum_{\forall t} \text{LOSS}(d_t, d'_t)$$

MSE for continuous attributes + **CE** for categorical attributes

SECTION 4

Experiments

The dataset, protocol, and metrics used to put the model to the test.

Dataset — a realistic, highly imbalanced log

Property	Value
Total transactions	~1,809,600
Fraudulent	~9,600 ($\approx$ 0.53%)
Node types	Customer, Merchant, Transaction
Key fields	cc_num, merchant_id, amount, time, geo
Label usage	calibration & evaluation only

Setup & why AUC-PR is the metric that matters

Setup

- Train on **legitimate-only** split
- Calibrate τ on held-out legit
- Baselines: **GraphSAGE**, **FI-GRL**

Metrics

- **AUC-PR** — key under imbalance
- **F1** — precision/recall at τ
- ROC-AUC — **optimistic** when negatives dominate

Caveat

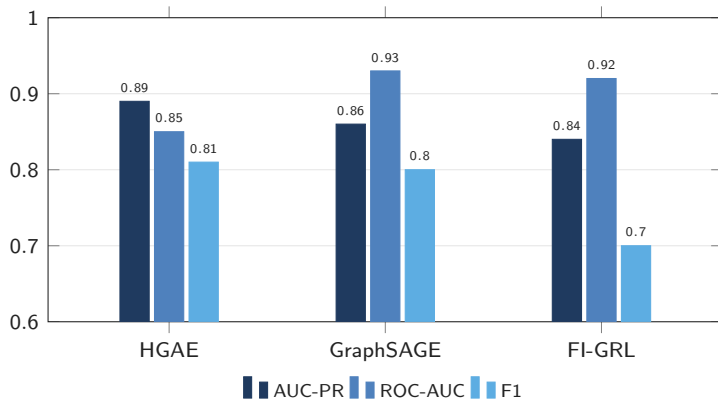
Reported encoder depth = **124 layers** — implausible under oversmoothing; blocks reproduction.

SECTION 5

Results

What the numbers reveal — and a critical reading of their limits.

Results — a precision–recall vs. ROC trade-off



HGAE leads on AUC-PR & F1 — but its ROC-AUC is the lowest.

Comparison — superiority is nuanced, not uniform

Method	AUC-PR	F1	ROC-AUC
HGAE	0.89	0.81	0.85
GraphSAGE	0.86	0.80	0.93
FI-GRL	0.84	0.70	0.92

Verdict

Wins where it matters under imbalance (**AUC-PR**, **F1**), but **loses on ROC-AUC** — a trade-off the original paper never reconciles.

Discussion — five formulation concerns

1. **Implausible depth**: 124 layers vs. oversmoothing $\rightarrow$ not reproducible
2. **Attribute-only decoding**: structure enters the score only *indirectly*
3. **Incomplete VAE**: no KL term $\rightarrow$ weak “variational” claim
4. **Overstated superiority**: lowest ROC-AUC, never reconciled
5. **Ambiguous threshold**: $\mu + 2\sigma$ vs. F1-sweep $\rightarrow$ possible leakage

Key takeaway

#2 is the deepest: it undercuts the paper's own “topology is what others miss”.

Limitations inherited from the GNN family

- **Scalability**: $O(nE)$ full-graph infeasible
- **Static-graph fallacy**: blind to velocity attacks
- **Heterophily**: fraud camouflages among legit
- **No explainability**: opaque score
- **Oversmoothing / -squashing** with depth
- **Latent collapse**: no contrastive term

These map back to the **challenges** raised at the start — the loop closes.

SECTION 6

Conclusion

Key takeaways and a research agenda for robust, label-free fraud detection.

Future work — every fix targets a named limitation

- **Masked & contrastive GAEs** (MGAE, IrGAE) → collapse + attribute-only
- **Architectural simplification** (linear prop., orthogonal init) → scale + depth
- **Temporal heterogeneous GNNs** → velocity / evolving fraud
- **Adaptive thresholds** → drop the global Gaussian assumption
- **Explainable attribution** → per-attribute loss, attention maps

- GAEs reframe fraud as **one-class anomaly detection** — beating label scarcity
- HGAE fuses **type-specific attention** + unsupervised reconstruction
- But superiority is **nuanced**; formulation & GNN limits remain
- Forward: **masked/contrastive, temporal, simplified, explainable**

Key takeaway

Learn what *normal* looks like on the graph and let fraud betray itself — but verify every claim honestly.

Thank you — Questions?

References

1. Majumder, S., Singh, M. T., Prasad, R. K., et al. *Heterogeneous Graph Auto-Encoder for Credit Card Fraud Detection*. International Journal of Computer Applications (IJCA), 32(2), 124–138, 2025.
2. Liu, Z., Chen, C., Yang, X., Zhou, J., Li, X., Song, L. *Heterogeneous Graph Neural Networks for Malicious Account Detection*. CIKM '18, 2077–2085, 2018.
3. Kipf, T. N., Welling, M. *Variational Graph Auto-Encoders*. arXiv:1611.07308, 2016.
4. Alon, U., Yahav, E. *On the Bottleneck of Graph Neural Networks and its Practical Implications*. ICLR, 2021.
5. Emergent Mind. *Graph Neural Network (GNN)-based Autoencoders*. Online resource, <https://www.emergentmind.com>, 2026 (updated Jan 2026).